

Evidence of Loyalty Signalling - A Costly but Necessary Strategy for Building Trust

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Abstract

In recent years, the political discourse of Hong Kong has undergone a visible "Mainlandization," characterized by the increasing adoption of *guanqiang* (officialese) and frequent references to Central Government initiatives such as "New Quality Productive Forces" and "Patriots Administering Hong Kong." This paper provides an empirical analysis of loyalty signaling, particularly in the speeches of Hong Kong Chief Executives (CEs) from 1997 to 2022, utilizing text mining techniques to quantify the prevalence of *guanqiang* and different classes of loyalty signalling strategies. We also examine the temporal evolution of semantic similarity of CE speeches relative to those of Mainland Chinese leaders. The findings reveal a distinct cyclical pattern: loyalty signaling intensifies following periods of socio-political friction or "trust crises" between Hong Kong and the Central Government. The research suggests that this discursive convergence is not merely a stylistic shift, but a reparative strategy—a tactical move designed to regain political confidence and signal institutional alignment during times of uncertainty.

Keywords: Loyalty Signalling, Guanqiang, Central-HK Relationship, Text Mining, Cosine Similarity

1. INTRODUCTION

Rho-class is specially designed for research articles, technical/lab reports, and academic documentation. This guide will help you understand the features and proper use of the template, enabling you to efficiently structure and format your documents with clarity and precision.

This class includes the following files in the rho-class folder:

- `rho.cls` — The main class file.
- `rhoenvs.sty` — A package containing custom environments.
- `rhobabel.sty` — A support package for multilingual.
- `README.md` — With essential information.

Additionally, this template includes a sample document `main.tex`, a bibliography file `rho.bib`, as well as figures, tables and code examples to help you get started quickly.

1.1. License

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2. METHODOLOGY

2.1. Data Collection

The data utilized in this project is organized according to the following directory structure:

- CE's speeches: `Data/Data_Raw/Speeches_and_Articles`
- Government press releases: `Data/Data_Raw/Press_Release`
- Central leaders' HK-related speeches: `Data/Data_Raw/Central/Liaison_Office/State_Leader`

The primary source of data consists of Chief Executive (CE) speeches and government press releases spanning from 1997 to 2026.¹

¹Press release (Traditional Chinese): <https://www.ceo.gov.hk/tc/press.html>;
CE speeches: <https://www.ceo.gov.hk/tc/speeches.html>; Archive of past CE's speeches and press releases: <https://www.ceo.gov.hk/tc/links.html>

Data regarding central leaders' speeches about Hong Kong was accessed from the Liaison Office of the Central People's Government (LOCPG) website: <http://www.locpg.gov.cn/zxzx/zysy.htm>

2.2. Data Cleaning Wrangling

The abstract and keywords are declared in the preamble of the document with `\begin{abstract}...\end{abstract}` environment alongside the dedicated `\keywords{}` command to allow the class file to automatically handle their formatting and positioning during the title creation process.

2.3. Keyword Detection

The `\rhostart{}` command, provides a personalized lettrine for the beginning of the first paragraph as shown in this example.

2.4. Cosine Similarity Analysis

This class includes a customized design for the table of contents, which is disabled by default. To enable it, insert the `\tableofcontents` command before your first section.

3. RESULTS AND FINDINGS

Rho-class provides shorthand commands for cross-referencing figures, tables, code listings, and equations. To customize the label format, edit the `rhobabel.sty` file. Label translations for Spanish are handled automatically.

- `\figref{}` — Uses the label style defined by `\figlabel`.
- `\figsref{}` — Uses the label style defined by `\figslabel`.
- `\tabref{}` — Uses the label style defined by `\tablabel`.
- `\coderef{}` — Uses the label style defined by `\captionlabelcode`.

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This document class was prepared on Overleaf and compiled with pdfTeX. No errors were found during the compilation process. Rho-class supports external editors, though additional setup may be required.

- `\eqref{}` — Uses the label style defined by `\eqlabel`.
- These commands generate fully hyperlinked references, both the label and the number, ensuring navigation to the referenced element when clicked in the PDF.

4. FIGURES

Fig.Fig. ?? shows a 3D surface visualization of a bivariate function.

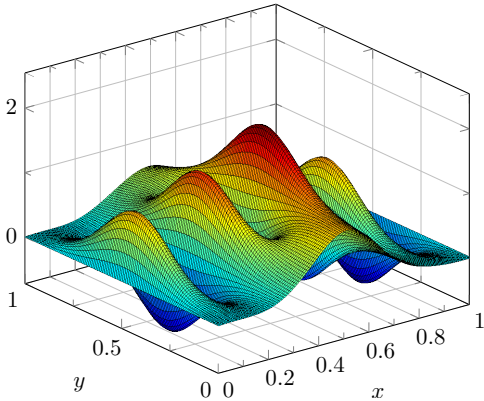


Figure 1. 3D surface obtained from PGFPlots [PFGPlots].

5. TABLES

TablaTable ?? shows an example table of some astronomical objects.

Table 1. Astronomical Object Data

Object	Type	Distance (Light Years)
Alpha Centauri	Star system	4.37
Betelgeuse	Red supergiant star	642.5
Andromeda	Spiral galaxy	2.537 million
Earth	Planet	0
Sirius	Binary star system	8.6

Note: The table contains data of some famous celestial objects.

The `\tabletext{}` command is used to add notes to tables easily.

6. DOCUMENT DATA

6.1. Front-Matter Footer Block*

A new block called `\docinfo{}`, has been added to the bottom right of the first page, acting as a floating element which stays fixed in place. This replaces the corresponding author block previously located above the abstract in earlier versions.

The `\docinfo{}` block was created to give you greater control over supplementary document data, such as publication dates, licensing information, extended author details, and other front-matter notes. You can fully customize its content to suit your needs.

Additionally, the new `\corres{}` command serves as a flexible alternative to `\thanks{}`. Since modifying the standard `\thanks{}` command can be complex and its positioning is often rigid, `\corres{}` was introduced to offer a more adaptable solution without sacrificing functionality.

NotaNote

You can use `\corres{}` as many times as needed since it automatically ensures that each entry remains correctly associated. For best results, I recommend using text symbols rather than mathematical ones (e.g., `\textasteriskcentered`, `\textdagger`, etc.).

If `\corres{}` is not used, `\docinfo{}` will automatically adjust its position. The implementation was designed to handle any combination of front-matter elements you might need when starting your document. In this example template, you'll find a minimal demonstration that combines both to illustrate their interaction.

6.2. Headers and Footers*

6.2.1. Headers

On all pages except the first, the document title and journal appear in the header. Since twoside layout is enabled, its position alternates depending on whether the page is odd or even.

6.2.2. Footers

In the first page, six elements display supplementary information for your document:

- `\journalname{}` — Displays the journal name, university, institute, company, or any institutional affiliation.
- `\theday{}` — Displays the current date.
- `\vol{}` — Specifies the volume number of the publication.
- `\no{}` — Specifies the issue number within the volume.
- `\journal{}` — Defines the full journal title.
- `\thepage` — Shows the current page number and the total count.

In all cases, you can freely rearrange or customize the order of these elements by modifying the class file to suit your needs. If any of them are not required, simply omit the corresponding command — the remaining elements will automatically adjust their spacing to remain evenly distributed.

7. RHO CUSTOM PACKAGES

7.1. Rhobabel.sty

This package have all the commands that automatically translate from English to Spanish when this custom package is defined.

By default, this document has its content in English. However, at the beginning of the document you will find a recommendation when writing in Spanish.

You may modify this package if you want to use other language than English or Spanish. This will make easier to translate your document without having to modify the class document.

7.2. Rhoenvs.sty

This package provides custom environments designed to enhance the visual presentation and structure of your document. Key examples include rhoenv, info, and note.

Their style is defined by `rhoenvstyle` — allowing you to customize their appearance according to your document's requirements.

An example using the `rhoenv` environment is shown below.

Custom Title

This is an example of the custom title environment. To add a title type this command `[frametitle=Custom Title]` next to the beginning of this environment (as shown in this example).

The `info` and `note` are environments which have a predefined title and translate their title to Spanish automatically when this language is defined.